

ANU Physics Academic Integrity Declaration Form

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Date: 28 sep 2026 Assessment Item: assignment 6

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You may not present material produced by generative AI as your own work, as this is an academic integrity breach. If you have used AI in this assessment, the use must be declared here:

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☐ **AI tools were used** as described below:

AI Tools Used: _____

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☐ Mathematical calculations ☐ Problem solving ☐ Code generation
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Chat Evidence Required - you must include your complete AI conversation(s):

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I declare that:

- ☒ This AI use declaration is complete and accurate
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- ☒ I understand the physics concepts and can explain my work
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Signature:  Date: 28 sep 2026

Assignment 6

September 26, 2026

1 Question 1

two masses, each of mass m , are connected by a spring with spring constant k .

(a)

since the two masses are identical, the centre of mass is in the middle of the spring. if one mass moves a distance x away from the centre of mass, the other mass moves a distance x in the opposite direction.

total change in the length of the spring is

$$x + x = 2x$$

from the hint, if the length of a spring is halved, its effective spring constant doubles. so, each half of the spring has spring constant

$$k_{\text{half}} = 2k$$

for one mass, the restoring force is $F = -2kx$
using Newton's second law,

$$F = m \frac{d^2x}{dt^2}$$

$$m \frac{d^2x}{dt^2} = -2kx$$

rearranging,

$$\frac{d^2x}{dt^2} + \frac{2k}{m}x = 0$$

the standard equation for simple harmonic motion is

$$\frac{d^2x}{dt^2} + \omega^2x = 0$$

comparing the two equations gives

$$\omega^2 = \frac{2k}{m}$$

therefore

$$\omega = \sqrt{\frac{2k}{m}}$$

since $\omega = 2\pi f$, we have

$$2\pi f = \sqrt{\frac{2k}{m}}$$

therefore, the resonant frequency is

$$f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

(b)

we are given $m = 1$ AMU and $f = 125$ THz. Converting the mass into SI units,

$$m = 1.66054 \times 10^{-27} \text{ kg}$$

$$f = 125 \times 10^{12} \text{ Hz} = 1.25 \times 10^{14} \text{ Hz}$$

from part (a),

$$f = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$$

multiplying both sides by 2π

$$2\pi f = \sqrt{\frac{2k}{m}}$$

squaring both sides,

$$4\pi^2 f^2 = \frac{2k}{m}$$

$$k = 2\pi^2 f^2 m$$

substituting the values,

$$k = 2\pi^2 (1.25 \times 10^{14})^2 (1.66054 \times 10^{-27})$$

$$k \approx 510.6 \text{ N/m}$$

hence,

$$k \approx 5.11 \times 10^2 \text{ N/m or } k \approx 511 \text{ N/m}$$

2 Question 2

we are given:

- $r = 0.05 \text{ kg/s}$
- $x_0 = 10 \text{ cm} = 0.1 \text{ m}$
- $v_0 = 0$
- $k = 5 \text{ N/m}$
- $m = 1 \text{ kg}$

(a)

the equation for a damped mass-spring system is

$$m \frac{d^2 x}{dt^2} + r \frac{dx}{dt} + kx = 0$$

substituting the given values,

$$\frac{d^2 x}{dt^2} + 0.05 \frac{dx}{dt} + 5x = 0$$

the standard form is

$$x'' + 2\gamma x' + \omega_0^2 x = 0$$

comparing coefficients,

$$2\gamma = 0.05$$

$$\gamma = 0.025 \text{ s}^{-1}$$

the undamped angular frequency is

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{5}{1}} = \sqrt{5} \approx 2.236 \text{ rad/s}$$

the damped angular frequency is

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2}$$

$$\omega_d = \sqrt{5 - (0.025)^2}$$

$$\omega_d \approx 2.23593 \text{ rad/s}$$

for an underdamped oscillator, the general solution is

$$x(t) = e^{-\gamma t} [A \cos(\omega_d t) + B \sin(\omega_d t)]$$

therefore,

$$x(t) = e^{-0.025t} [A \cos(2.23593t) + B \sin(2.23593t)]$$

using the initial displacement,

$$x(0) = 0.10$$

at $t = 0$,

$$0.1 = A \cos(0) + B \sin(0)$$

since $\cos(0) = 1$ and $\sin(0) = 0$, we get

$$A = 0.10$$

therefore,

$$x(t) = e^{-0.025t} [0.10 \cos(2.23593t) + B \sin(2.23593t)]$$

differentiating to apply the initial velocity condition,

$$v(t) = -\gamma e^{-\gamma t} [A \cos(\omega_d t) + B \sin(\omega_d t)] + e^{-\gamma t} [-A\omega_d \sin(\omega_d t) + B\omega_d \cos(\omega_d t)]$$

at $t = 0$,

$$v(0) = -\gamma A + B\omega_d$$

since $v(0) = 0$,

$$0 = -\gamma A + B\omega_d$$

$$B = \frac{\gamma A}{\omega_d}$$

substituting the values,

$$B = \frac{(0.025)(0.10)}{2.23593}$$

$$B \approx 0.001119$$

therefore, the position as a function of time is

$$x(t) = e^{-0.025t} [0.10 \cos(2.23593t) + 0.001119 \sin(2.23593t)] \text{ m}$$

(b)

initially, the mass is displaced by $x_0 = 0.10$ m and has zero velocity.
therefore, the initial energy is entirely spring potential energy

$$E_0 = \frac{1}{2} k x_0^2.$$

substituting,

$$E_0 = \frac{1}{2} (5) (0.10)^2$$

thus,

$$E_0 = \frac{1}{2} (5) (0.01)$$

and

$$E_0 = 0.025 \text{ J.}$$

the damped angular frequency is

$$\omega_d = 2.23593 \text{ rad/s}$$

therefore, the oscillation period is

$$T = \frac{2\pi}{\omega_d}.$$

substituting,

$$T = \frac{2\pi}{2.23593} \approx 2.8101 \text{ s}$$

after 10 periods,

$$t = 10T$$

so

$$t = 10(2.8101) \approx 28.101 \text{ s}$$

the amplitude decays according to

$$A(t) = A_0 e^{-\gamma t}$$

since the energy is proportional to the square of the amplitude,

$$E(t) = E_0 e^{-2\gamma t}$$

therefore,

$$E(t) = 0.025 e^{-2(0.025)(28.101)}$$

calculating the exponent,

$$2(0.025)(28.101) = 1.40505$$

thus,

$$E(t) = 0.025 e^{-1.40505}.$$

this gives

$$E(t) \approx 0.006134 \text{ J}$$

therefore, the energy lost is

$$\Delta E = E_0 - E(t)$$

hence,

$$\Delta E = 0.025 - 0.006134$$

$$\boxed{\Delta E \approx 0.01887 \text{ J}}$$

3 Question 3

the pipe is closed at one end and open at the other. at the closed end there is a displacement node, while at the open end there is a displacement antinode. for the fundamental mode, the pipe contains one quarter of a wavelength:

$$L = \frac{\lambda_1}{4}$$

$$\lambda_1 = 4L$$

using $f = \frac{c}{\lambda}$,

$$f_1 = \frac{c}{4L}$$

for higher modes, only odd harmonics are allowed. therefore,

$$n = 1, 3, 5, 7, \dots$$

$$L = \frac{n\lambda_n}{4}$$

$$4L = n\lambda_n,$$

$$\lambda_n = \frac{4L}{n}$$

we have,

$$f_n = \frac{c}{\lambda_n}$$

substituting for λ_n ,

$$f_n = \frac{c}{4L/n}$$

hence,

$f_n = \frac{nc}{4L}, \quad n = 1, 3, 5, 7, \dots$
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4 Question 4

we have,

- speed of dolphin, $v_s = 40 \text{ km/h} = 40 \left(\frac{1000}{3600} \right) = 11.111 \text{ m/s}$
- frequency produced by sound, $f_s = 1000 \text{ Hz}$

the Doppler-shifted frequency is

$$f_{\text{heard}} = f_s \frac{v}{v - v_s},$$

where v is the speed of sound in the medium.

(a)

the speed of sound in water is approximately $v = 1500$ m/s. therefore,

$$f_{\text{heard}} = 1000 \left(\frac{1500}{1500 - 11.11} \right)$$
$$f_{\text{heard}} \approx 1007.5 \text{ Hz}$$

therefore,

$$f_{\text{heard}} \approx 1008 \text{ Hz}$$

(b)

taking the speed of sound in air to be approximately $v = 343$ m/s and substituting in $f_{\text{heard}} = f_s \frac{v}{v - v_s}$,

$$f_{\text{heard}} = 1000 \left(\frac{343}{343 - 11.11} \right)$$
$$f_{\text{heard}} \approx 1033.5 \text{ Hz}$$

therefore,

$$f_{\text{heard}} \approx 1034 \text{ Hz}$$